Bálint Gyevnár

Carnegie Mellon University
Institute for Complex Social Dynamics
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Employment

09/2025 – present **Carnegie Mellon University**
Postdoctoral Research Associate with Nihar Shah and Atoosa Kasirzadeh
Current projects: assessment of author understanding at grecaptcha.com; investigating failure modes of AI scientist systems, such as indirect data poisoning and backdooring;

Education

09/2021 – 05/2026 **University of Edinburgh**
Ph.D. in Natural Language Processing
Supervisors: Stefano Albrecht, Shay Cohen, Chris Lucas
Thesis: Action Explanation of Multi-Agent Systems via Counterfactual Reasoning [link]
Examiners: Ann Nowé, Bonan Zhao

09/2016 – 05/2021 **University of Edinburgh**
M.Inf in Informatics
Supervisor: Maria Wolters
Thesis: Comparison of Account Survival on Twitter During the First Wave of COVID-19

08/2018 – 06/2019 **Nanyang Technological University**
Exchange Year in Computer Science

Past Employment

07/2023 – 12/2024 **University of Edinburgh**, Edinburgh Futures Institute
Research Assistant with Shannon Vallor, Atoosa Kasirzadeh

09/2020 – 09/2025 **University of Edinburgh**, School of Informatics
Teaching Assistant (see *Teaching*)

05/2020 – 10/2020 **FiveAI**, Edinburgh
Research Intern with Stefano Albrecht

Awards

2024 **Colours Award for Outstanding Volunteering Contribution to Sports**
Edinburgh University Sports Union

2023 **AI100 Early Career Essay Competition, Featured Essay**
One Hundred Year Study on Artificial Intelligence, Stanford University

2022 **Trustworthy Autonomous Systems Early Career Researcher Award**
UK Research & Innovation £4,000

2021 **Shape the Future of ITS Competition Award**
IEEE Intelligent Transportation Systems Society $1,000

Preprints

Google Scholar: https://scholar.google.com/citations?user=fLyES3oAAAAJ
Semantic Scholar: https://www.semanticscholar.org/author/Balint-Gyevnar/2007920131
* denotes equal contribution.

[P4] **Bálint Gyevnár** and Simon DeDeo. Autonomous AI Agents Systematically Diverge From Human Preferences in Proof Formalization. *Submitted to Nat. Mach. Intell., 2026.* [link]

[P3] Justin Payan*, **Bálint Gyevnár***, Atoosa Kasirzadeh, and Nihar B. Shah. greCAPTCHA: Assessing Understanding as Evidence of Research Authorship Under Generative AI. *Under review at CHI, 2026.* [link]

[P2] **Bálint Gyevnár**, Atoosa Kasirzadeh, and Nihar B. Shah. Distributed Denial of Science: How Indirect Data Poisoning of AI Systems Can Industrialize Scientific Fraud. *Submitted to PNAS, 2026.* [link]

[P1] **Bálint Gyevnár** and Atoosa Kasirzadeh. Bridging the Gap in the Responsible AI Divides. *Under revision at Patterns, 2026.* [link]

Peer-Reviewed Publications

Journal Publications

[J4] **Bálint Gyevnár*** and Atoosa Kasirzadeh*. AI Safety for Everyone. *Nature Machine Intelligence, 2025.* [link]

[J3] Anton Kuznietsov*, **Bálint Gyevnár***, Cheng Wang, Steven Peters, and Stefano. V. Albrecht. Explainable AI for Safe and Trustworthy Autonomous Driving: A Systematic Review. *IEEE Transactions on Intelligent Transportation Systems, 2024.* [link]

[J2] **Bálint Gyevnár***, Gautier Dagan*, Coleman Haley*, Shangmin Guo*, and Frank Mollica. Communicative Efficiency or Iconic Learning: Do Communicative and Acquisition Pressures Interact to Shape Colour-Naming Systems? *Entropy, 2022.* [link]

[J1] Ibrahim H. Ahmed, Cillian Brewitt, Ignacio Carlucho, Filippos Christianos, Mhairi Dunion, Elliot Fosong, Samuel Garcin, Shangmin Guo, **Bálint Gyevnár**, Trevor McInroe, Georgios Papoudakis, Arrasy Rahman, Lukas Schäfer, Massimiliano Tamborski, Giuseppe Vecchio, Cheng Wang, and Stefano V. Albrecht. Deep Reinforcement Learning for Multi-Agent Interaction. *AI Communications, 2022.* [link]

Conference Publications

[C6] **Bálint Gyevnár**, Christopher G. Lucas, Stefano V. Albrecht, and Shay B. Cohen. Integrating Counterfactual Simulations with Language Models for Explaining Multi-Agent Behaviour. *AAMAS 2026.* [link]

[C5] **Bálint Gyevnár**, Stephanie Droop, Tadeg Quillien, Shay B. Cohen, Neil R. Bramley, Christopher G. Lucas, and Stefano V. Albrecht. People Attribute Purpose to Autonomous Vehicles When Explaining Their Behavior: Insights From Cognitive Science for Explainable AI. *CHI 2025.* [link]

[C4] **Bálint Gyevnár**, Cheng Wang, Christopher G. Lucas, Shay B. Cohen, and Stefano V. Albrecht. Causal Explanations for Sequential Decision-Making in Multi-Agent Systems. *AAMAS 2024.* [link]

[C3] **Bálint Gyevnár**, Nick Ferguson, and Burkhard Schafer. Bridging the Transparency Gap: What Can Explainable AI Learn From the AI Act? *ECAI 2023.* [link]

[C2] Cillian Brewitt, **Bálint Gyevnár**, Samuel Garcin, and Stefano V. Albrecht. GRIT: Fast, Interpretable, and Verifiable Goal Recognition with Learned Decision Trees for Autonomous Driving. *IROS 2021.* [link]

[C1] Stefano V. Albrecht, Cillian Brewitt, John Wilhelm, **Bálint Gyevnár**, Francisco Eiras, Mihai Dobre, and Subramanian Ramamoorthy. Interpretable Goal-based Prediction and Planning for Autonomous Driving. *ICRA 2021.* [link]

Workshop Publications

[W2] **Bálint Gyevnár**[*] and Mark Towers[*]. Objective Metrics for Human-Subjects Evaluation in Explainable Reinforcement Learning. *RLDM 2025.* [link]

[W1] **Bálint Gyevnár**, Massimiliano Tamborski, Cheng Wang, Christopher G. Lucas, Shay B. Cohen, and Stefano V. Albrecht. A Human-Centric Method for Generating Causal Explanations in Natural Language for Autonomous Vehicle Motion Planning. *IJCAI 2022 Workshop on Artificial Intelligence for Autonomous Driving.* **Best paper runner-up.** [link]

Teaching

Tutor and Demonstrator (University of Edinburgh)

Spring 2024 EFIE11173: Evaluating Sustainable Lands and Cities
EFIE11179: Data, Mobility, Infrastructure

Spring 2024 INFR11010: Reinforcement Learning

Autumn 2023 INFR11194: Doing Research in Natural Language Processing

Spring 2022 INFR11202: Machine Learning Theory

Autumn 2020 INFR08027: Informatics 2C - Introduction to Computer Systems
INFR10069: Introductory Applied Machine Learning.

Mentoring

09/2026–present **Kerissa Duliga**; Master's student of Computer Science

09/2026–present **Yiming Shao**; Undergraduate student of Computer Science

10/2026–present **Brandon Zhou**; Master's student of Philosophy

Service

10/2025 **Workshop Organiser**
Evaluating Explainable AI and Complex Decision-Making, ECAI 2025

Reviewer / PC Member
Journals: Nature Communications (2025); Artificial Intelligence Review (2025); International Journal of Human-Computer Interaction (2026); IEEE Transactions on Intelligent Transportation Systems (2025); Journal of Autonomous Agents and Multi-Agent Systems (2026)
Conferences: AAAI (2026); AIES (2025); CHI (2025, 2026); ECAI (2023); FAccT (2026, 2027); ICRA (2025); IASEAI (2026, 2027); IROS (2025); NeurIPS (2025); XAI (2026)
Workshops: Academic Mindtrek (2026), EMNLP BlackboxNLP (2025), NeurIPS AI4Metascience (2026)

Media Coverage

07/2026 Nature News. Coverage of *Distributed Denial of Science*. [link]
Doctored data sets could trick AI agents

07/2026 El País. Coverage of *Distributed Denial of Science*. [link]
El problema de los clips ya está aquí: las IA pueden destruir el mundo sin querer

09/2026 Times Higher Education. Frances Jones. Coverage of *greCAPTCHA*. [link]
Researchers submitting AI-assisted papers they "don't understand"

07/2025 Qubit. Coverage of *AI Safety for Everyone*. [link]
Igazságkereső AI helyett MuskGPT lett a milliárdos chatbotjából, a Grokból

04/2024 AI Hub. Coverage of *Causal Explanations for Multi-Agent Systems*. [link]
Interview with Bálint Gyevnár: Creating explanations for AI-based decision-making systems

Invited Talks

forthcoming **Imperical College London**. Digital Foundry Seminar Series. (Hosted by Eleonora Giunchiglia)

forthcoming **UK AISI**. Seminar Series. (Hosted by Christopher Summerfield)

forthcoming **DeepMind**. Seminar Series. (Hosted by Atoosa Kasirzadeh)

forthcoming **University of Oxford**. Synthetic Society Lab. (Hosted by Luc Rocher)

forthcoming **University of Cambridge**. Language Technology Lab. (Hosted by Anna Korhonen)

forthcoming **University of Cambridge**. TRACE Lab. (Hosted by Umang Bhatt)

forthcoming **University of Edinburgh**. ILCC/CDT NLP Seminar Series. (Hosted by Alex Taylor)

06/2026 **Columbia University**. Machine Learning Summer School. (Hosted by Gary Kazantsev)

03/2026 **IVADO**. Workshop on Social Reasoning and the Ecology of Thought. (Hosted by James Evans)

06/2025 **University of California, Berkeley**. 9th Workshop of the Center for Human-Compatible AI

07/2024 **Charles University and Czech Technical University**. Machine Learning and Modelling Seminar. (Hosted by Martin Holeňa)